INCH-POUND

MIL-B-52251C(ME)
6 December 1988
SUPERSEDING
MIL-B-52251B(ME)
2 September 1980

MILITARY SPECIFICATION

BRACKET, SLING ATTACHMENT: FOR BEAMS AND GIRDERS OF LINE OF COMMUNICATION BRIDGES

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lifting bracket to be bolted to the beams or girders used in the construction of the line of communication bridges.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420
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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATIN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A108 - Standard Quality Cold-Finished Carbon Steel Bars.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The sling bracket shall be as shown in figure 1 and as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Steel. Steel shall be flat bar steel conforming to ASTM A108, UNS G 10350.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the bracket may be newly fabricated from recovered materials to the maximum extent practicable, provided the bracket produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the bracket.

3.4 Identification marking. The sling bracket shall be metal-die stamped as shown on the figure. Die stamping shall be legible after painting.

3.5 Treatment and painting. The sling bracket shall be cleaned, treated, and painted in accordance with MIL-T-704, type A except that the color number designation shall be 34087 in accordance with FED-STD-595.

3.6 Workmanship.

3.6.1 Metal fabrication. Metal used in fabrication shall be free from scale, fins, cracks, sharp bends and kinks. The straightening of material shall be done by methods that will not cause injury to the material. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. The corners and edges shall be rounded to provide a 1/16-inch minimum radius to permit safe handling by personnel. All burrs and projections shall be removed from holes and edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article. The first article sling bracket shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The sling bracket shall be examined as specified herein for the following defects:

Major:

101. Nonconformance to the dimensions shown on the figure.
102. Material not as specified.
103. Workmanship not as specified.
104. Steel not as specified.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the bracket.

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Minor:

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

106. Containers not as specified for level A, B or C. Each incorrect container shall be considered one defect.
107. Quantities packed together exceed the limitations specified for level A, B or C.
108. Strapping not zinc coated for level A.
109. Making missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation is not necessary for the sling brackets as they are completely painted.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The sling brackets shall be packed together in a close fitting box conforming to PPP-B-601, overseas type, style optional, grade B, in quantities not to exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.2.2 Level B. The sling brackets shall be packed together as specified in 5.2.1 except the box shall be domestic type and strapping shall not be required to be zinc coated. As an alternate the sling brackets may be packed together in a close fitting box conforming to PPP-B-636, V3c, style optional. The gross weight or size of the box shall not exceed the limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Sling brackets shall be packed together in close fitting boxes conforming to PPP-B-636, class domestic. The gross weight and size of the boxes shall not exceed the limitations imposed by the box specification. Box closure shall be in accordance with the box specification.

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5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sling attachment bracket is intended for use as a means of engaging sling hooks for lifting beams and girders.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article model (see 3.2).
d. Level of packing required (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a first article model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Beams and girders
Bracket
Sling attachment for beams and girders

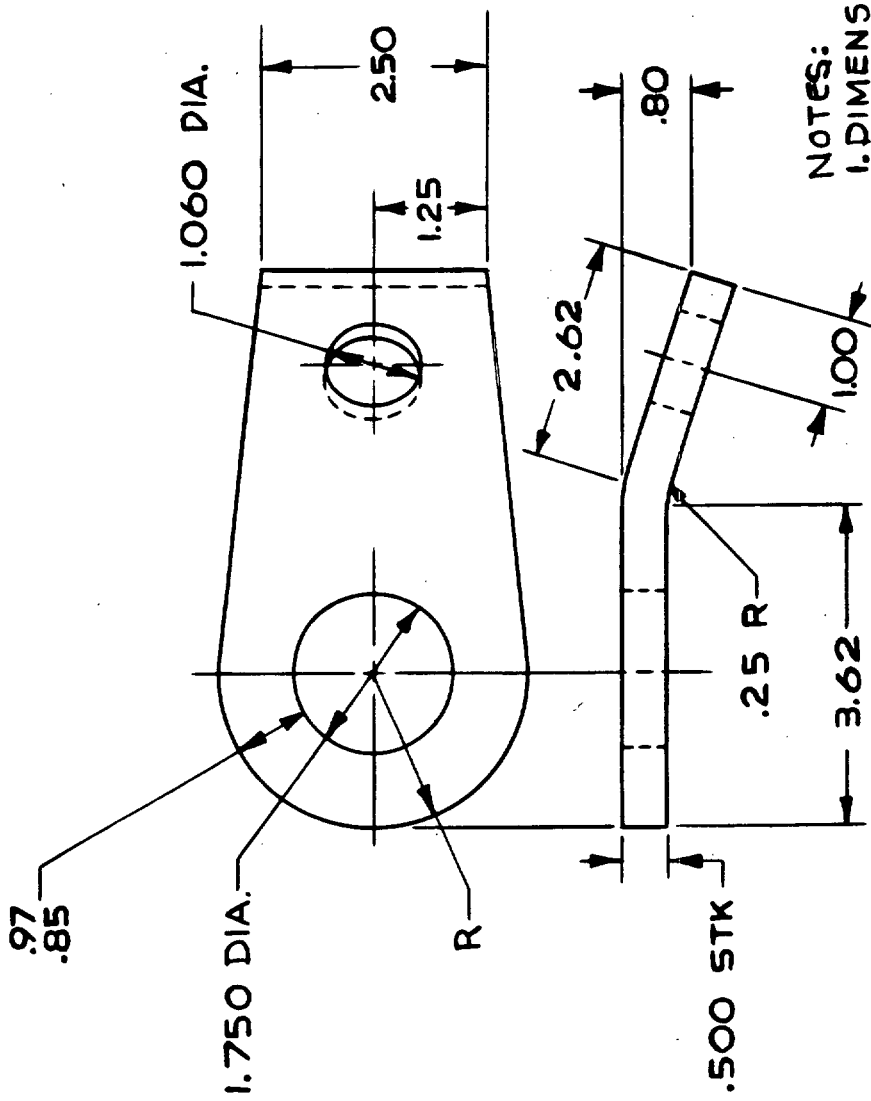
6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project No. 5420-A362

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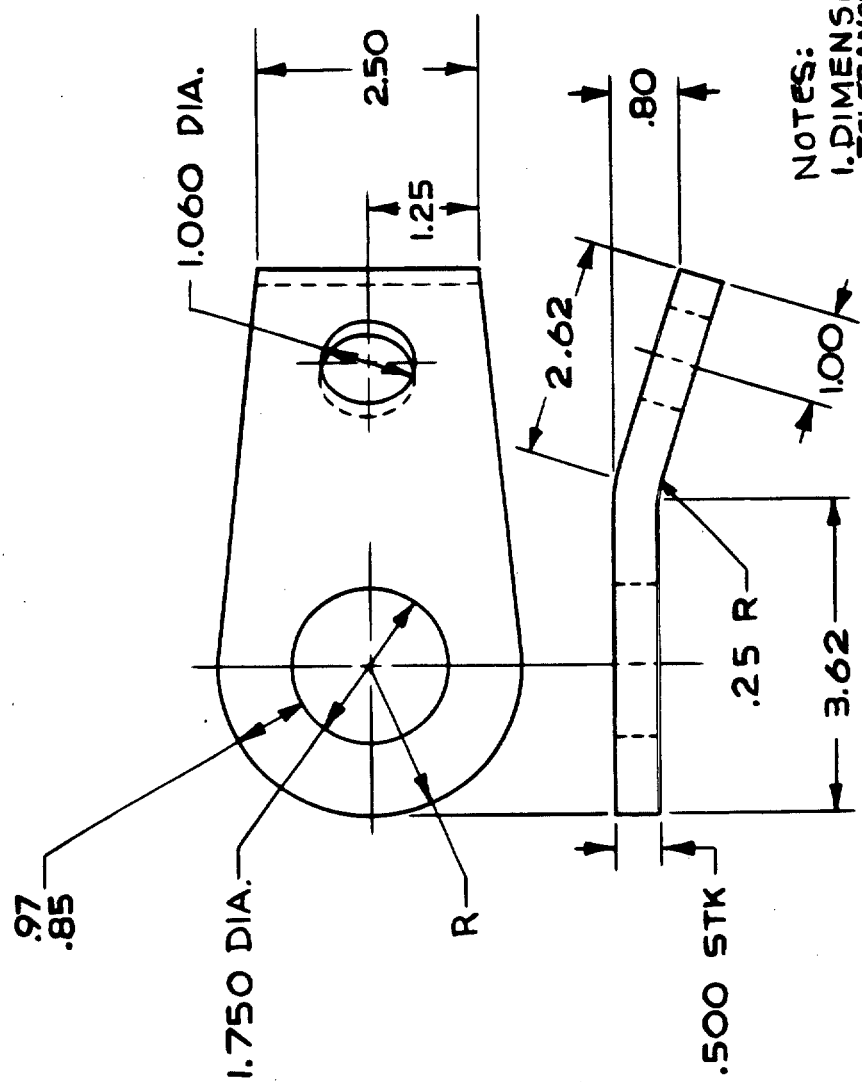


NOTES:
1. DIMENSIONS ARE IN INCHES, TOLERANCES XX ±.06 XXX ±.030
2. METAL STAMP WITH 1/8" HIGH CHARACTERS:
5420-00-764-5643
BRACKET
CONTR. NO
MFR. IDENT. U.S

FIGURE 1. SLING ATTACHMENT BRACKET FOR BEAM AND GIRDER BRIDGES

X-2352 B

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FIGURE 1. SLING ATTACHMENT BRACKET FOR BEAM AND GIRDER BRIDGES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-52251C(ME)	Bracket, Sling Attachment: For Beams and Girders of Line of Communication Bridges

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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